

Mental Health Agent

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Mental Health Support is Hard to Access

- 1 in 5 adults experience mental health issues every year: yet many never receive treatment
- People often turn to:
 - Google – overwhelming / unreliable
 - Chatbots – generic, unsafe, not personalized
- Early support can reduce severity, prevent crises, and encourage professional help

Our Solution (Mental Health Agent)

An **agentic AI system** that:

- Provides empathetic conversation
- Retrieves real research (PubMed)
- Detects crisis situations BEFORE the LLM
- Personalizes responses using a User Profile

System Architecture

Key Components:

- Safety Service (pre-LLM)
- LangGraph Agent
- MCP Tool Server
- User Profile System
- Frontend (React + SSE)

High-Level Flow



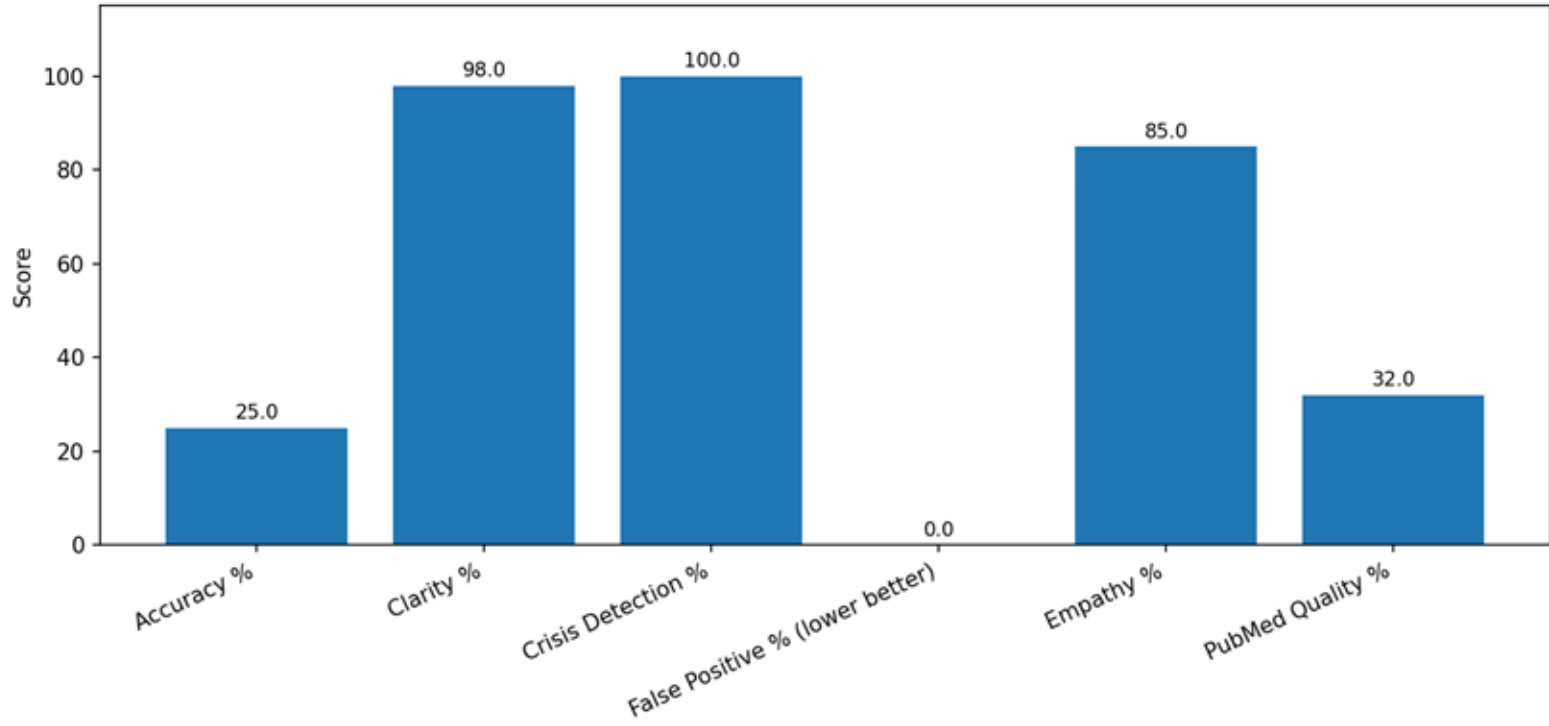
Key Technical Choices

Component	Choice	Why
LLM	Claude	Strong reasoning + tool use
Agent	LangGraph	Structured loop control
Tools	MCP	Modular + decoupled
Backend	FastAPI	Async + streaming
Frontend	React	Real-time UX

Evaluation Benchmarks/Datasets

- MedQA (https://huggingface.co/datasets/bigbio/med_qa) – Correctness
- HealthSearchQA (<https://huggingface.co/datasets/katielink/healthsearchqa>) – Patient Clarity
- Safety Prompts (custom-made) – Safety

Mental Health Bot Evaluation Scores



Demo

Link: <https://youtu.be/8z2vJnoWSJ4>

Lessons Learned

1. **Agentic > Prompt Engineering**

- Letting the model choose tools is powerful

2. **Safety Must Be External**

- LLM alone is not enough

3. **Personalization is Critical**

- Biggest jump in user experience quality

4. **Tool Design Matters**

- Clear schemas → better decisions

Future Improvements

Technical

- PostgreSQL for persistent profiles
- Auth system (with UVA emails)
- Full streaming tool-use

Product

- Mood tracking over time
- Personalized therapy plans
- Role-specific views (students, counselors, admins)

Research

- Benchmark against mental health datasets
- A/B test personalization impact

Key Takeaways

This project demonstrates how to build a **safe, personalized, agentic AI system**, not just a chatbot.

Core Principles:

- Safety first
- Tool grounding
- User-aware responses
- Modular architecture

Division of Responsibilities

Tammy Ngo

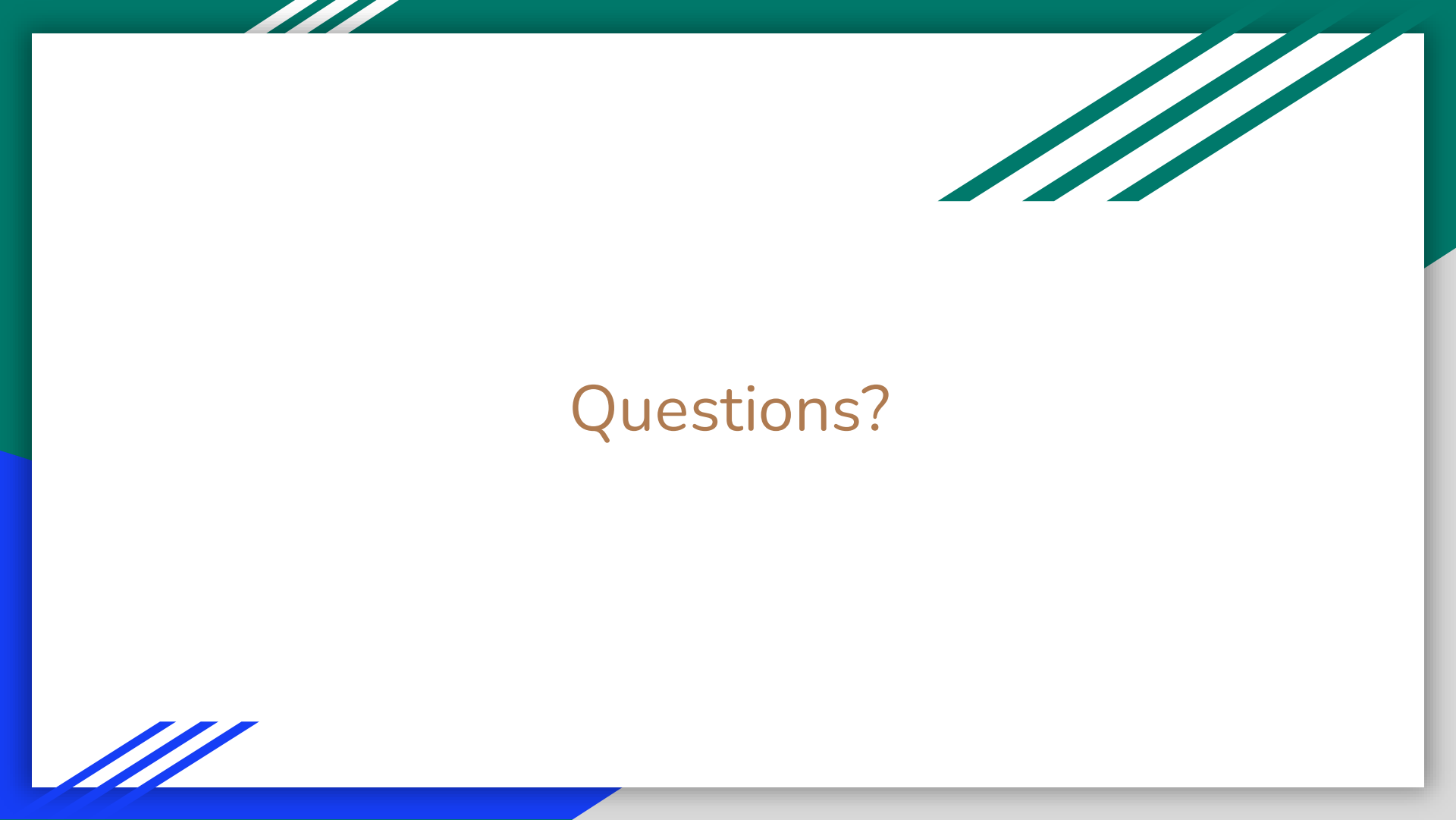
- Implemented Mental Health Triage
- Upgraded to Safety Layer v2.0
- Implemented User Profile

Matt Juntima

- Refracted framework for MCP-compatibility
- Implemented LangGraph
- Updated UI
- Upgraded to Safety Layer v2.0
- Setup public url for agent
- Dockerized project
- Recorded Demo

Sebastian Pop

- Upgraded Claude system prompt
- Added Clinical Trials tool
- Implemented tool routing logic
- Ran evaluations for model



Questions?